Mathematical Modelling — Tutorial 1

How these tutorials work. Each tutorial has pen-and-paper parts and a *computational task* that you carry out in whatever language you prefer (MATLAB is recommended; Python or Julia are equally fine). The point of the code is not the language, but it is that a classmate can regenerate your figure from your folder. Each week you add a folder `week-NN/<username>/` to the course tutorials repository on GitHub, containing a half-page `report.md`, the figure(s) requested, the code that produced them, and a short `README` giving the command to regenerate them. You submit this as a pull request and review one classmate's in return (see the repository's `CONTRIBUTING.md`). Parts marked *Challenge* are optional, for the motivated or when time allows. Chapter and appendix references are to the course notes. Every `report.md` should also close with one sentence of *interpretation*: what does your result mean for the real system you modelled, and not merely as a piece of mathematics?

Pre-tutorial/question 0, or something like that : create a github account if you don't have one, see ..... for instructions if you don't know it. Search for the course organization? I have the link, should I provide it? how are they accept it to it? or should they send me an email with their github username? Explain to me, but also add instructions. If you haven't ever used Github I recommend watching ..... or reading ..... we will do a short introduction to the necessary git commands in the first tutorial, but you will need to be able to create a repository, clone it, and push your changes.

Tutorial 1 (Week 1 — Dimensional analysis in practice)

Notes: Chapter on dimensional analysis. This week also serves as your onboarding to the course repository.

1. Rederive, by dimensional analysis alone, the period of a simple pendulum $T = \sqrt{L/g}\,\Phi(\theta_0)$, stating clearly what the method fixes and what it leaves in the unknown function Φ.

2. Pick one further system — the frequency of a mass on a spring, the speed of deep-water waves, or the draining time of a tank — and carry out the same analysis: list the variables, form the dimensionless groups, and state precisely what is predicted and what is left undetermined.

3. A rigid-body (compound) pendulum — an extended rigid object swinging about a pivot — has small-amplitude period $T = 2\pi\sqrt{L_{\text{eff}}/g}$, where g is gravity and $L_{\text{eff}} = I/(mR)$ is its *effective length*, built from the body's mass m, its moment of inertia I about the pivot, and the pivot-to-centre-of-mass distance R (see the notes, Chapter 6). Without computing any moment of inertia, explain why dimensional analysis already guarantees the $\sqrt{L_{\text{eff}}/g}$ dependence.

Computational task. Taylor's blast-wave argument predicts $R = C\,(Et^2/\rho)^{1/5}$ for the radius R of a blast at time t. Generate a synthetic "photographic" data set: fix $\rho = 1.2$ and take the constant $C = 1$ (its theoretical value), choose an energy E, compute $R(t)$ at a dozen times, and add a few percent of random noise to mimic measurement error. Now *pretend you do not know* E: fit a straight line to $\log R$ versus $\log t$, check that the slope is close to $2/5$, and recover E from the intercept (with $C = 1$ and ρ known, the intercept gives E/ρ). Report how close your estimate is to the true E.

Repo deliverable. 1. Your fitting code, 2. a log–log plot showing the data and the fitted line, and 3. a `report.md` stating the recovered slope and energy — closing with what this shows: that a handful of photographs, plus dimensional analysis alone, suffice to weigh an explosion. This is your first pull request — get the workflow working now, while the mathematics is easy. (how do we assess the "review part"? need to decide on times as well. Tutorials are on Thursdays, but people may not be able to finish it, therefore the reviewing part will take time? how do I mark? I will also check the code, randomly?)